
* STAUD.Pro
* Version 2006 Bld 1005.US
* Proprietary Program of
* Research Engineers, Intl.
* Date= AUG 14, 2007
* Time= 14:34:20
*
* USER ID: Meinhardt Infrastructure Pte Ltd
*

1. STAAD SPACE PAINT BOOTH DUCT SUPPORT

INPUT FILE: Paint Booth Duct Support.STD
2. START JOB INFORMATION
3. JOB NAME - DUBAI METRO PROJECT
4. JOB CLIENT - GOVERNMENT OF DUBAI - ROADS & TRANSPORT AUTHORITY
5. ENGINEER DATE AUG-07
6. END JOB INFORMATION
7. INPUT WIDTH 79
8. UNIT SYSTEM
9. UNIT CONVERSIONS
10. 11 0.0 0.0 0.0 12 0.9 0.0 0.0 13 0.9 0.0 0.9 14 0.0 0.0 0.9
11 0.0 1.5 0.0 22 0.9 1.5 0.9 23 0.9 1.5 0.9 24 0.0 1.5 0.9
12 0.0 3.0 0.0 32 0.9 3.0 0.9 33 0.9 3.0 0.9 34 0.0 3.0 0.9
13 0.0 4.5 0.0 42 0.9 4.5 0.9 43 0.9 4.5 0.9 44 0.0 4.5 0.9
14 0.0 6.0 0.0 52 0.9 6.0 0.9 53 0.9 6.0 0.9 54 0.0 6.0 0.9
15 0.0 7.5 0.0 62 0.9 7.5 0.9 63 0.9 7.5 0.9 64 0.0 7.5 0.9
16 0.0 9.0 0.0 72 0.9 9.0 0.9 73 0.9 9.0 0.9 74 0.0 9.0 0.9
17 0.0 10.5 0.0 82 0.9 10.5 0.9 83 0.9 10.5 0.9 84 0.0 10.5 0.9
18 0.0 12.0 0.0 92 0.9 12.0 0.9 93 0.9 12.0 0.9 94 0.0 12.0 0.9
19 0.0 13.5 0.0 102 0.9 13.5 0.9 103 0.9 13.5 0.9 104 0.0 13.5 0.9
20 0.0 15.0 0.0 112 0.9 15.0 0.9 113 0.9 15.0 0.9 114 0.0 15.0 0.9
21. MEMBER INCIDENCES
22. 21 41 22 33; 24 24 21; 31 31 32 33; 34 34 31
23. 21 41 22 33; 24 24 21; 31 31 32 33; 34 34 31
24. 61 62 63; 64 64 61; 71 71 72 73; 74 74 71
25. 81 82 83; 84 84 81; 91 91 92 93; 94 94 91
26. 101 101 102 103; 104 104 101; 111 111 112 113; 114 114 111
27. 1001 11 21 1004; 2001 21 31 2004; 3001 31 41 3004; 4001 41 51 4004
28. 5001 51 61 5004; 6001 61 71 6004; 7001 71 81 7004; 8001 81 91 8004
29. 9001 91 101 9004; 10001 101 111 10004
30. 1101 11 1102 11 34; 1103 31 44; 1104 41 54; 1105 51 64
31. 1106 61 74; 1107 71 84; 1108 81 94; 1109 91 104; 1110 101 114
32. 1201 12 23; 1202 22 33; 1203 32 43; 1204 42 53; 1205 52 63
33. 1206 62 73; 1207 72 83; 1208 82 93; 1209 92 103; 1210 102 113
34. 1301 24 13; 1302 24 33; 1303 44 33; 1304 44 53; 1305 64 53
35. 1306 64 73; 1307 84 73; 1308 84 93; 1309 104 93; 1310 104 113
36. MEMBER RELEASES
37. 1101 TO 1110 1201 TO 1210 1301 TO 1310 BOTH MX MY MZ
38. MEMBER ENDSON 201 TO 1210 1301 TO 1310
39. *1101 TO 1110 1201 TO 1210 1301 TO 1310
40. MEMBER PROPERTY BRITISH
41. MEMBER PROPERTY BRITISH
42. MEMBER PROPERTY BRITISH
43. MEMBER PROPERTY BRITISH
44. MEMBER PROPERTY BRITISH
45. MEMBER PROPERTY BRITISH
46. MEMBER PROPERTY BRITISH
47. MEMBER PROPERTY BRITISH
48. MEMBER PROPERTY BRITISH
49. MEMBER PROPERTY BRITISH
50. MEMBER PROPERTY BRITISH
51. MEMBER PROPERTY BRITISH
52. MEMBER PROPERTY BRITISH
53. MEMBER PROPERTY BRITISH
54. MEMBER PROPERTY BRITISH
55. MEMBER PROPERTY BRITISH
56. MEMBER PROPERTY BRITISH
57. MEMBER PROPERTY BRITISH
58. MEMBER PROPERTY BRITISH
59. MEMBER PROPERTY BRITISH
60. MEMBER PROPERTY BRITISH
61. MEMBER PROPERTY BRITISH
62. MEMBER PROPERTY BRITISH
63. MEMBER PROPERTY BRITISH
64. MEMBER PROPERTY BRITISH
65. MEMBER PROPERTY BRITISH

PAINT BOOTH DUCT SUPPORT
21 TO 24 31 TO 34 41 TO 44 51 TO 54 61 TO 64 71 TO 74 81 TO 84 -
42. 91 TO 94 101 TO 104 111 TO 114 TABLE ST TUB1001005.0
43. 1001 TO 1004 2001 TO 2004 3001 TO 3004 4001 TO 4004 5001 TO 5004 -
44. 6001 TO 6004 7001 TO 7004 8001 TO 8004 9001 TO 9004 -
45. 10001 TO 10004 TABLE ST TUB2002005.0
46. 1101 TO 1110 1201 TO 1210 1301 TO 1310 -
47. TABLE ST UABOX8DX6
48. DEFINE MATERIAL START
49. ISOTROPIC STEEL
50. E 2.05E+008
51. POISSON 0.3
52. DENSITY 7850
53. ALPHA 1.2E-005
54. RHO 0.00785
55. END DEFINE MATERIAL
56. CONSTANTS
57. MATERIAL STEEL ALL
58. SUPPORTS
59. 11 12 13 14 PINNED
60. LOAD 1 SELFWEIGHT
61. SELFWEIGHT Y -1
62. LOAD 2 SDL
63. MEMBER LOAD
64. 21 23 31 33 41 43 51 53 61 63 71 73 81 83 -
65. 91 93 101 103 111 113 CON GY -0.32

PAINT BOOTH DUCT SUPPORT
21 TO 24 31 TO 34 41 TO 44 51 TO 54 61 TO 64 71 TO 74 81 TO 84 -
42. 91 TO 94 101 TO 104 111 TO 114 TABLE ST TUB1001005.0
43. 1001 TO 1004 2001 TO 2004 3001 TO 3004 4001 TO 4004 5001 TO 5004 -
44. 6001 TO 6004 7001 TO 7004 8001 TO 8004 9001 TO 9004 -
45. 10001 TO 10004 TABLE ST TUB2002005.0
46. 1101 TO 1110 1201 TO 1210 1301 TO 1310 -
47. TABLE ST UABOX8DX6
48. DEFINE MATERIAL START
49. ISOTROPIC STEEL
50. E 2.05E+008
51. POISSON 0.3
52. DENSITY 7850
53. ALPHA 1.2E-005
54. RHO 0.00785
55. END DEFINE MATERIAL
56. CONSTANTS
57. MATERIAL STEEL ALL
58. SUPPORTS
59. 11 12 13 14 PINNED
60. LOAD 1 SELFWEIGHT
61. SELFWEIGHT Y -1
62. LOAD 2 SDL
63. MEMBER LOAD
64. 21 23 31 33 41 43 51 53 61 63 71 73 81 83 -
65. 91 93 101 103 111 113 CON GY -0.32

66. LOAD 3 LL
67. MEMBER LOAD
68. 21 23 31 33 41 43 51 53 61 63 71 73 81 83 -
69. 91 93 101 103 111 113 CON GY -1.0
70. LOAD 11 WL1
71. MEMBER LOAD
72. *WL = 0.7KPA
73. 1001 1002 2001 2002 3001 3002 4001 4002 5001 5002 6001 6002 -
74. 7001 7002 8001 8002 9001 9002 10001 10002 UNI GX -0.32
75. LOAD 12 WL2
76. MEMBER LOAD
77. *WL = 0.7KPA
78. 1001 1004 2001 2004 3001 3004 4001 4004 5001 5004 6001 6004 -
79. 7001 7004 8001 8004 9001 9004 10001 10004 UNI GX -0.32
80. LOAD 21 EQ1 (EARTHQUAKE LOAD)
81. JOINT LOAD
82. 21 22 31 32 41 42 51 52 61 62 71 72 81 82 91 92 -
83. 101 102 111 112 EZ -0.71
84. LOAD 22 EQ1 (EARTHQUAKE LOAD)
85. JOINT LOAD
86. 21 24 31 34 41 44 51 54 61 64 71 74 81 84 91 94 -
87. 101 104 111 114 EX -0.71
88. LOAD COMB 101 1.4DL + 1.6LL
89. LOAD COMB 101 1.4DL + 1.6LL
90. LOAD COMB 101 1.2DL + 1.2LL + 1.2WL1
91. LOAD COMB 101 1.2DL + 1.2LL + 1.2WL1
92. LOAD COMB 101 1.2DL + 1.2LL + 1.2WL1
93. LOAD COMB 101 1.2DL + 1.2LL + 1.2WL1
94. LOAD COMB 101 1.2DL + 1.2LL + 1.2WL1
95. 1.4 2 1.4 1.1 1.4
96. LOAD COMB 1211 1.0DL + 1.4WL1
97. 1.0 2 1.0 1.1 1.4
98. LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
99. PAINT BOOTH DUCT SUPPORT
99. 1.1 2 1.2 3 1.2 12 12
100. LOAD COMB 1112 1.4DL + 1.4WL2
101. 1.1 2 1.4 1.2 1.4
102. LOAD COMB 1212 1.0DL + 1.4WL2
103. 1.1 2 1.0 1.2 1.4
104. LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ1
105. LOAD COMB 1022 1.2DL + 0.5LL + 1.0EQ2
106. 1.1 2 1.2 3 0.5 21 -1.0
107. LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ1
108. 1.1 2 1.2 3 0.5 21 -1.0
109. LOAD COMB 1122 1.2DL + 0.5LL - 1.0EQ2
110. 1.1 2 1.2 3 0.5 21 -1.0
111. LOAD COMB 1122 1.2DL + 0.5LL - 1.0EQ2
112. 1.1 2 1.2 3 0.5 21 -1.0
113. LOAD COMB 1221 0.9DL + 1.0EQ1
114. 1.0 2 0.9 21 1.0
115. LOAD COMB 1222 0.9DL + 1.0EQ2
116. 1.0 2 0.9 21 1.0
117. LOAD COMB 1321 0.9DL - 1.0EQ1
118. 1.0 2 0.9 21 -1.0
119. LOAD COMB 1322 0.9DL - 1.0EQ2
120. 1.0 2 0.9 21 -1.0
121. *SL5 COMB 2001 1.0DL + 1.0LL
122. LOAD COMB 2001 1.0DL + 1.0LL
123. *SL5 COMB 2002 1.0DL + 0.5LL
124. LOAD COMB 2002 1.0DL + 0.5LL
125. 1.0 2 1.0 3 0.5
126. 1.0 2 1.0 3 0.5
127. LOAD COMB 2011 DL + LL + WL1
128. 1.0 2 1.0 3 0.5 11 1.0
129. LOAD COMB 2012 DL + LL + WL2
130. 1.0 2 1.0 3 0.5 12 1.0
131. PERFORM ANALYSIS
132. PERFORM ANALYSIS
133. PERFORM ANALYSIS

PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBER/ELEMENTS/SUPPORTS = 44/ 110/ 4
ORIGINAL/FINAL BAND-WIDTH= 36/36
TOTAL MEMBER CASES = 7, TOTAL DEGREES OF FREEDOM = 252
NUMBER OF STIFFNESS MATRIX = 10 DOUBLE
REQD/AVAIL. DISK SPACE = 12.3/ 345694.1 MB

134. LOAD LIST 2001 2002 2011 2012
135. PRINT JOINT DISPLACEMENTS ALL
136. PRINT JOINT DISPLACEMENTS ALL
137. PRINT BOOTH DUCT SUPPORT